

Madhura Kale

📍 Pune, Maharashtra | ✉ madhurakale0807@gmail.com | ☎ +91-9607029819 | 🔗 madhura-kale-979833291 | 🌐 Madhura0807

Summary

Computer Engineering undergraduate (2027 batch) with hands-on experience across data analytics, machine learning, and full-stack development. Skilled in Python, SQL, Power BI, and end-to-end model development, with a published peer-reviewed research paper in deep learning.

Education

AISSMS Institute of Information Technology <i>B.Tech. in Computer Engineering CGPA: 7.34/10 (approx. 73.4%)</i>	Aug 2023 – Jun 2027 Pune, India
MP International School and Junior College <i>Higher Secondary Education 73.67%</i>	Jun 2021 – Mar 2023 Pune, India
Kendriya Vidyalaya Ganeshkhind <i>Secondary Education 90%</i>	Jun 2020 – Mar 2021 Pune, India

Projects

Customer Churn Analysis and Prediction | *Python, Pandas, NumPy, SQL, Power BI, Scikit-learn, XGBoost*

- Analyzed customer, subscription, and transaction datasets using Python, Pandas, NumPy, and SQL, performing data cleaning, EDA, feature engineering, and KPI analysis to surface churn drivers and revenue patterns.
- Designed an interactive Power BI dashboard using data modeling, DAX measures, slicers, and drill-downs to track churn rate, retention, revenue, and high-risk customer segments.
- Developed and compared Logistic Regression, Random Forest, and XGBoost classifiers to predict churn, evaluating performance using precision, recall, F1-score, and ROC-AUC.

Legal Document Reader | *Python, FastAPI, RAG, FAISS, Groq LLM, MongoDB, React.js*

- Built a Retrieval-Augmented Generation (RAG) application enabling natural-language querying of legal documents using semantic search and LLM-based responses.
- Implemented OCR, text chunking, FastEmbed embeddings, and FAISS vector search to retrieve relevant legal information; developed FastAPI REST APIs with MongoDB for storage and query history.
- Built a React/Vite frontend for document upload and legal Q&A, delivering a complete full-stack solution.

Transaction Intelligence Engine | *Python, NLP, RapidFuzz, Groq LLM*

- Developed a Python system to process unstructured transaction descriptions, identifying merchant names, categories, and confidence scores.
- Combined rule-based matching, regex, fuzzy matching, and LLM-based classification with caching and batch processing for efficient, accurate transaction classification.

Publication

Deep Learning-Based Web App for Plant Leaf Disease Classification Using MobileNetV2 | *IJCRT*

- Published a peer-reviewed research paper presenting a MobileNetV2 transfer learning framework for automated plant leaf disease classification.
- Developed and deployed a web application classifying plant leaf diseases from uploaded images using the trained model.

Technical Skills

Languages: Python, C++, SQL

Data & Analytics: Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Feature Engineering, KPI Analysis, Data Visualization, Power BI, DAX, Excel

Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Transfer Learning, Model Training, Model Evaluation, TensorFlow

Generative AI: LLMs, RAG, LangChain, Prompt Engineering, Embedding Generation, Vector Databases (FAISS)

Backend & APIs: FastAPI, Flask, Streamlit, Postman

Databases: MySQL, MongoDB

Cloud: AWS Fundamentals

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks

Developer Tools: Git, GitHub, Version Control, VS Code

Achievements

Leadership & Ownership: Took end-to-end ownership of 8+ technical and cultural events as Secretary of iConnect across two semesters, coordinating cross-functional teams and delivering events for 100+ attendees.

Problem Solving: Solved 200+ Data Structures and Algorithms problems on LeetCode across Easy, Medium, and Hard difficulty levels.

Hackathon: Advanced past Round 1 of the Advitya Hackathon at IIT Bombay by presenting a business solution to a judging panel.